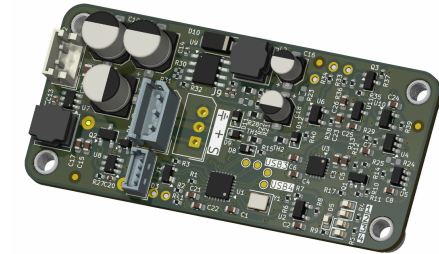


"elec_hub_and_dyn" is a part of the Grabette/Gripette project, it is a under-HAT for rasp zero 2w with a USB hub and a Dynamixel device.



Release	Date	Designer	Check	Comments
A1	16/04/2026	EB	SN	Initial drawing
B1	01/06/2026	EB	SN	[fix] Spacers in the BoM [fix] DCDC is now 5A capable [fix] USB pogo-pins position [update] 5V DCDC is on Vbus [update] 4 mm spacers/pogo-pins [added] Test points + access to UART-Dynamixel [added] Hub reset rasp-accessible

- la rasp zero 2w fait convertisseur module camera vers usb
- on ajoute un converti USB-232 pour aller taper dans les moteurs de gripette
- un ajoute un hub pour factoriser tout ce petit monde sur un seul câble
- on met un DCDC 24-→12
- le tout sur un "under-HAT", sous la rasp, histoire de profiter de la paire diff USB sans utiliser de connecteur (juste des pogo-pins)
- universel: ça doit fonctionner sur OpenArm, S0100...
- options: IMU, connecteur Qwiic...

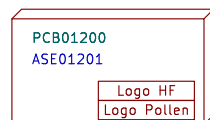
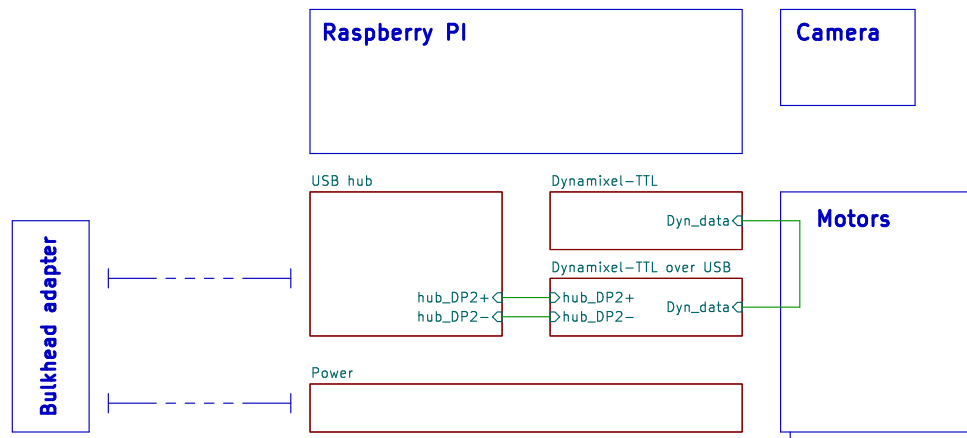
- S010x: USB connector + Dynamixel/Feotech @7,4V
- OpenArm: USB connector + 24V on XT60

Supplying the motors:

- through Vbatt (from the XT60): R28 must be populated
- through Dxl: R28 must be removed.

Motor communication:

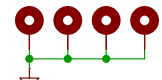
- USB (Dynamixel-TTL converter): R15 must be populated
- Dynamixel: R15 must be removed.



Fiducials



M2,5 x 4mm spacer



Pollen Robotics/HF

Sheet: /

File: elec_gripette_hub_and_dyn.kicad_sch

Title: Gripette – USB hub and Dynamixel device

Size: A4

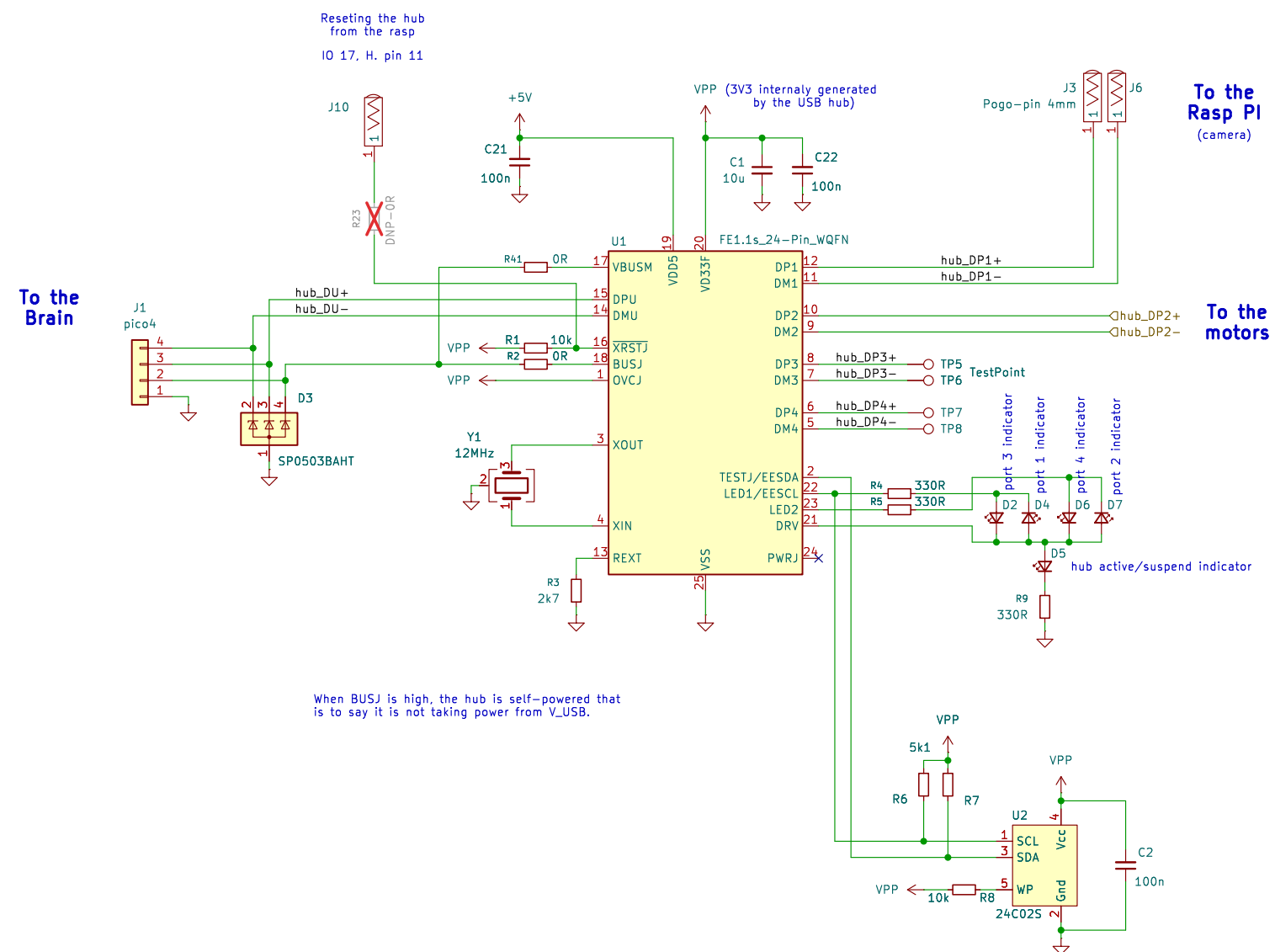
Date: 2026-05-27

Rev: B1

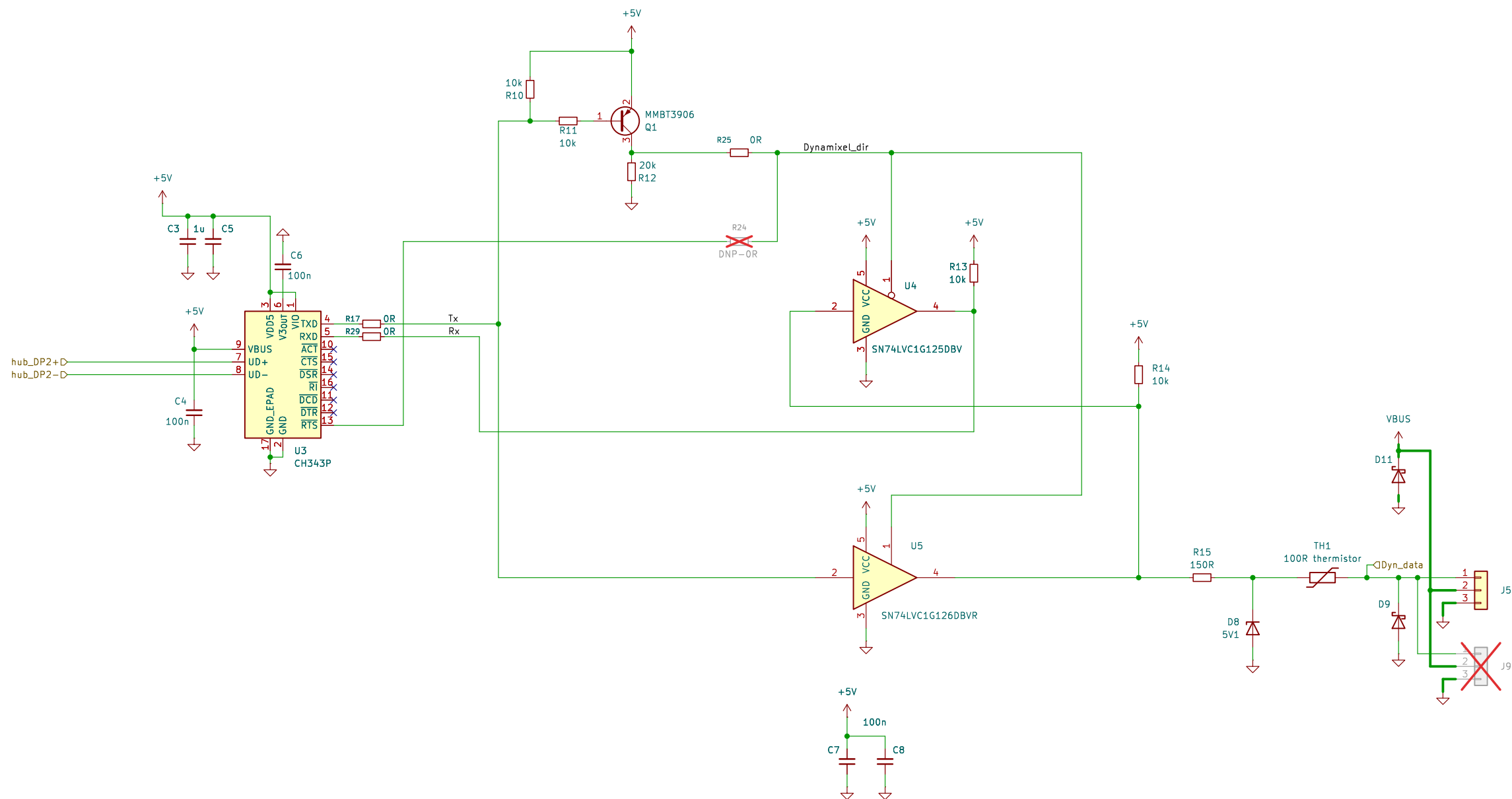
KiCad E.D.A. 9.0.9

Id: 1/5

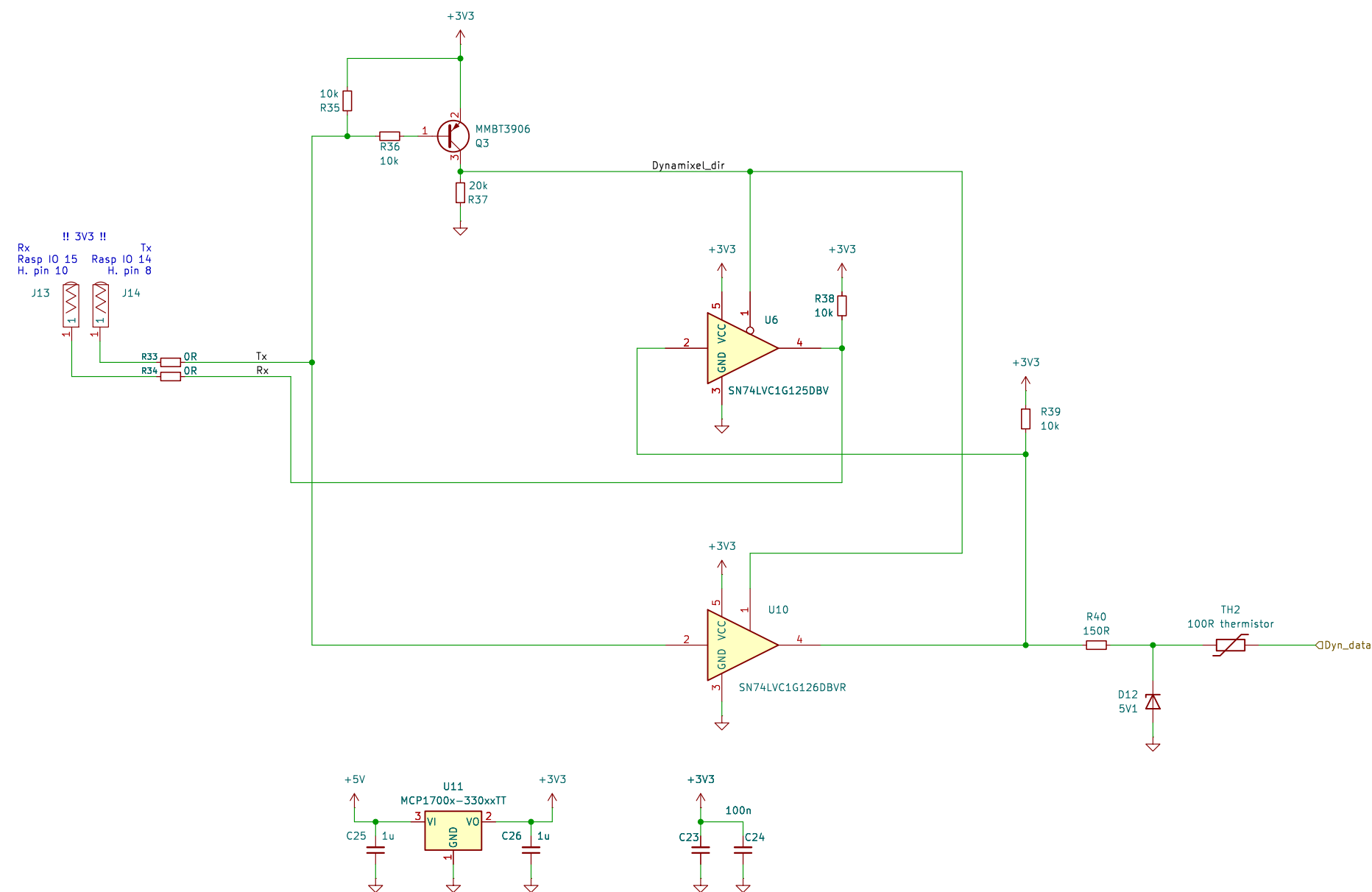
USB 2.0 hub



Dynamixel TTL



Dynamixel TTL
from Rasp.



DYNAMIXEL

